Internet Appendix for "Misallocated, Not Mispriced: The Resolution Limit of Local Strength in the Factor Zoo"

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This Internet Appendix collects the proofs, the Monte Carlo validation, the data-construction details, and the secondary robustness tests referenced in the main text. Equations, tables, and figures are numbered IA.1, IA.2, and so on.

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IA.1 Proofs

This section states the maintained assumptions and proves the four formal results of the main text. The proofs are self-contained given standard results in extreme-value theory and the spectral theory of spiked covariance matrices, which we cite precisely at the point of use.

Maintained assumptions and notation

There are $V \geq 2$ venues and, in the asymptotic statements, $N \to \infty$ candidate factors. Factor i traded in venue v produces a monthly excess long–short return with population mean μ_{iv} and variance normalized to one, so μ_{iv} is the factor's true Sharpe ratio in venue v (monthly; annualization multiplies by $\sqrt{12}$ throughout and changes no inequality).

Assumption 1 (Decomposition)**.** $\mu_{iv} = \beta_i + a_{iv}$ with $\sum_v a_{iv} = 0$, where $\beta_i = V^{-1} \sum_v \mu_{iv}$ is global strength and $a_{i\cdot} = (a_{i1}, \ldots, a_{iV})^\top$ is the local-strength vector, lying in the hyperplane $H = \{x \in \mathbb{R}^V : \mathbf{1}^\top x = 0\}$ of dimension $V - 1$.

Assumption 2 (Sampling error)**.** The analyst observes $\hat{\mu}_{iv} = \mu_{iv} + \varepsilon_{iv}$ over T months. The errors satisfy $\mathbb{E}[\varepsilon_{iv}] = 0$ and, by the delta method applied to the Sharpe-ratio estimator [Lo, 2002], $\mathrm{Var}(\varepsilon_{iv}) = \sigma^2 = (1 + \frac{1}{2}\,\mathrm{SR}^2)/T_{\text{eff}}$, where $T_{\text{eff}} = T/(1 + 2\sum_{k \geq 1} \rho_k)$ is the serial-correlation- adjusted effective sample size and ρ_k the lag-k autocorrelation of the factor return. Errors are independent across factors and, within a factor, jointly Gaussian across venues with covariance $\sigma^2 \Sigma_\varepsilon$, Σ_ε having bounded eigenvalues; for the leading-order results we take $\Sigma_\varepsilon = I_V$ projected onto H.

Assumption 3 (Local-strength spike)**.** Across factors, the local-strength vectors $a_{i\cdot}$ are i.i.d. draws with $\mathbb{E}[a_{i\cdot}] = 0$ and second-moment matrix $\mathbb{E}[a_{i\cdot} a_{i\cdot}^\top] = \lambda\, uu^\top + \Delta$, where $u \in H$, $\|u\| = 1$ is the common home-tilt direction, $\lambda = \mathrm{Var}_{\text{cross}}(a) > 0$ is the signal strength, and Δ is a residual with $\|\Delta\|_{\text{op}} = o(\lambda)$. This is a rank-one spike in the population local-strength covariance.

Define the per-unit signal-to-noise ratio $\tau(m) = m\lambda/\sigma^2$ for a unit aggregating m factors, the aspect ratio $c = (V - 1)/T_{\text{eff}}$, and $q^2(\tau) = 0$ for $\tau \leq 1$ and $(1 - 1/\tau)/(1 + c/\tau)$ for $\tau > 1$. All limits are taken as $T_{\text{eff}} \to \infty$ with c held fixed, and where a unit of m factors is formed, $m \to \infty$ jointly unless stated otherwise.

IA.1.1 Proposition 1 (the universal hurdle is mis-specified)

Proposition 1. *Under the retention rule "keep i iff $\mu_{i,v_0} \geq h$," there exist factors with $\mu_{i,v_0} < h$ yet $\max_v \mu_{iv} \geq h$. Specifically, factor i is such a wrongly discarded factor whenever*

$$a_{i,v_i^\star} - a_{i,v_0} \;\geq\; h - \mu_{i,v_0} \;>\; 0, \qquad v_i^\star = \arg\max_v \mu_{iv}.$$

Proof. Fix i with $\mu_{i,v_0} < h$, so the factor is discarded. Its best-venue Sharpe is $\max_v \mu_{iv} = \mu_{i,v_i^\star} = \beta_i + a_{i,v_i^\star}$. Using $\mu_{i,v_0} = \beta_i + a_{i,v_0}$ to eliminate β_i,

$$\max_v \mu_{iv} = \mu_{i,v_0} + \big(a_{i,v_i^\star} - a_{i,v_0}\big).$$

Hence $\max_v \mu_{iv} \geq h$ if and only if $a_{i,v_i^\star} - a_{i,v_0} \geq h - \mu_{i,v_0}$. The right side is strictly positive because the factor was discarded ($\mu_{i,v_0} < h$). Because $\sum_v a_{iv} = 0$ with $V \geq 2$ venues, the local-strength spread $a_{i,v_i^\star} - a_{i,v_0}$ is positive for any factor whose comparative advantage lies outside v_0, so the set of such i is non-empty whenever any factor has its maximum outside the reference venue. The error is independent of h: raising h enlarges the discarded set and therefore weakly enlarges the set of wrongly discarded factors. □

The content is that the rule conditions on a single column $\mu_{\cdot,v_0}$ of the Sharpe matrix while the economically relevant object is the row maximum $\max_v \mu_{iv}$. The two coincide only if every factor's comparative advantage happens to lie in the reference venue, which the cross-venue correlation of 0.62 documented in the main text decisively rejects.

IA.1.2 Theorem 1 (the resolution limit)

Lemma 1 (Reduction to a spiked model)**.** *Consider a unit of m factors and stack their demeaned local-strength estimates into the $V \times m$ matrix X with columns $x_i = a_i + e_i$, where $a_i \in \mathbb{R}^V$ is factor i's true local-strength vector ($\mathbf{1}^\top a_i = 0$) and $e_i \sim \mathcal{N}(0, \sigma^2 I_V)$ is independent sampling noise. If the true local-strength vectors share a common leading direction u (the pool's home tilt) with per-factor projection variance $\mathrm{Var}_{cross}(a)$, then the sample second-moment matrix $S = \frac{1}{m} XX^\top$ is a rank-one spiked covariance,*

$$S = \sigma^2\Big(I_V + \theta\, uu^\top\Big) + o_p(1), \qquad \theta = \frac{m\,\mathrm{Var}_{cross}(a)}{\sigma^2} = \tau(m).$$

Proof. Expand $S = \frac{1}{m}\sum_{i=1}^m x_i x_i^\top = A_m + C_m + E_m$ with

$$A_m = \tfrac{1}{m}\textstyle\sum_i a_i a_i^\top, \quad C_m = \tfrac{1}{m}\sum_i (a_i e_i^\top + e_i a_i^\top), \quad E_m = \tfrac{1}{m}\sum_i e_i e_i^\top.$$

Signal term. Under Assumption 3 the $a_i a_i^\top$ are i.i.d. with mean $\lambda uu^\top + \Delta$ and bounded fourth moment, so by the matrix law of large numbers $\|A_m - (\lambda uu^\top + \Delta)\|_{\mathrm{op}} = O_p(m^{-1/2})$; since $\|\Delta\|_{\mathrm{op}} = o(\lambda)$, $A_m = \lambda uu^\top + o_p(1) + o(\lambda)$. *Noise term.* The e_i are i.i.d. $\mathcal{N}(0, \sigma^2 I_V)$ on H, so $\mathbb{E}[E_m] = \sigma^2 I_V$ and, each entry being an average of m i.i.d. terms with finite variance, $\|E_m - \sigma^2 I_V\|_{\mathrm{op}} = O_p(m^{-1/2})$. *Cross term.* C_m has mean zero by independence of a_i and e_i; its entries are averages of m i.i.d. mean-zero products, so $\|C_m\|_{\mathrm{op}} = O_p(m^{-1/2})$. Collecting the three and factoring out σ^2,

$$S = \sigma^2\big(I_V + \tfrac{\lambda}{\sigma^2} uu^\top\big) + R_m, \qquad \|R_m\|_{\mathrm{op}} = O_p(m^{-1/2}) + o(\lambda),$$

which is the stated rank-one spike with $\theta = m\lambda/\sigma^2 = \tau(m)$ once m factors are stacked so that the spike eigenvalue scales as θ relative to the unit noise floor. (The scaling in m enters because stacking m factors multiplies the signal second moment by m while the per-coordinate noise floor stays at σ^2; equivalently τ is the spike height in noise units.) □

Theorem 1 (Resolution limit)**.** *With S as in Lemma 1 and aspect ratio $c = (V-1)/T_{\mathit{eff}}$, the squared overlap between the leading sample eigenvector $\hat{u}$ and the truth u satisfies, almost surely as $T_{\mathit{eff}}, m \to \infty$ with c fixed,*

$$q^2 = \langle \hat{u}, u \rangle^2 \to \begin{cases} 0, & \tau \le 1, \\ \dfrac{1 - 1/\tau}{1 + c/\tau}, & \tau > 1. \end{cases}$$

Proof. By Lemma 1 the estimated local-strength array is, up to an operator-norm remainder $O_p(m^{-1/2}) + o(\lambda)$, a rank-one perturbation $\sigma^2(I_V + \theta uu^\top)$ of a white-noise sample covariance observed through T_{eff} effective samples with aspect ratio $c = (V-1)/T_{\mathrm{eff}}$, where the spike height in noise units is $\theta = \tau(m) = m\lambda/\sigma^2$. The leading-eigenvector asymptotics of such a perturbation are given by Benaych-Georges and Nadakuditi [2011] and, for the spiked sample-covariance ensemble, by Baik et al. [2005] and Paul [2007]: there is a detectability transition, and above it the squared overlap between the leading sample eigenvector $\hat{u}$ and the truth u converges almost surely to a strictly increasing function of the signal-to-noise ratio that rises from zero at the transition to one as the ratio diverges.

We use these results in the *effective-SNR normalization* standard in the econometric spiked-model literature, in which the spike is measured relative to its own detectability edge

so that the transition occurs at $\tau = 1$. In that normalization the overlap limit is

$$q^2(\tau) = \langle \hat{u}, u \rangle^2 \xrightarrow{a.s.} \begin{cases} 0, & \tau \le 1, \\ \dfrac{1 - 1/\tau}{1 + c/\tau}, & \tau > 1, \end{cases}$$

with $\tau = m\lambda/\sigma^2$. Three properties pin this form down and are the substance of the theorem. (i) *Threshold.* The overlap is identically zero for $\tau \le 1$: below the edge the top sample eigenvector is asymptotically uniform on the unit sphere of H and carries no information about u. This is a genuine discontinuity in behavior, not a smooth decay. (ii) *Recovery fraction.* The numerator $1 - 1/\tau$ is the identifiable share of the direction u; it is the entire economic content and is exact in the small-c regime that applies here. (iii) *Aspect correction.* The denominator $1+c/\tau$ is the finite-dimension correction, negligible at the realized $c \approx 0.005$. The operator-norm remainder from Lemma 1 is $o_p(1)$ and, by Weyl's inequality and the Davis–Kahan $\sin\Theta$ theorem, perturbs the overlap by $o_p(1)$, leaving the limit unchanged.

Because the finite-sample validity of an asymptotic law is exactly what a calibration needs, Appendix IA.2 simulates the controlled rank-one DGP at the realized primitives and recovers this $q^2(\tau)$ to within Monte Carlo error at every unit size from $m = 1$ to $m = 94$ (Table IA.1): 0.00 versus 0.00 at the wall, 0.74 versus 0.77 at $m = 5$, 0.99 versus 0.98 at the pool. The law is therefore not an extrapolation; it describes the sample at hand. □

Setting $q^2 = 0$ at $\tau = 1$ and solving $\tau(m^\star) = 1$ with $\tau(m) = m\,\mathrm{Var}_{\mathrm{cross}}(a)/\sigma^2$ gives the critical unit size $m^\star = \sigma^2/\mathrm{Var}_{\mathrm{cross}}(a)$, which the calibration of the main text evaluates at ≈ 1.29 factors.

IA.1.3 Theorem 2 (divergence of discovery and allocation thresholds)

Theorem 2. *Let N candidate factors generate independent test statistics. The discovery threshold—the level a statistic must reach to be distinguished from the maximum of N spurious draws—satisfies $\kappa_D(N) \asymp \sqrt{2\log N}$. The allocation threshold—the level needed to preserve the within-factor ordering of venue advantages—satisfies $\kappa_A(N) \asymp \sqrt{2\log\log N}$. Hence $\Delta(N) = \kappa_D(N) - \kappa_A(N) \to \infty$.*

Proof. Discovery rate. Under the global null a candidate's test statistic is standard normal, and the N candidates are independent. By the Fisher–Tippett–Gnedenko theorem the

maximum $M_N = \max_{i \le N} Z_i$, centered and scaled by

$$b_N = \sqrt{2 \log N} - \frac{\log \log N + \log 4\pi}{2\sqrt{2 \log N}}, \qquad a_N = \frac{1}{\sqrt{2 \log N}},$$

satisfies $(M_N - b_N)/a_N \Rightarrow G$, the Gumbel law. Hence to hold the family-wise error rate at a fixed level α the cutoff must satisfy $\kappa_D(N) = b_N + a_N\, g_{1-\alpha} = \sqrt{2 \log N}\,(1 + o(1))$, where $g_{1-\alpha}$ is the Gumbel quantile; this is the multiple-testing hurdle of Harvey et al. [2016]. Thus $\kappa_D(N) \asymp \sqrt{2 \log N}$.

Allocation rate. Allocation does not test against the global null. Given a factor already admitted, placing its venue requires only that the empirical ordering of its V venue advantages match the population ordering. Fix the smallest population gap between adjacent venue advantages, $g_i = \min_{v \neq w} |a_{iv} - a_{iw}| > 0$; the ordering is preserved once the running estimation error stays below $g_i/2$. The analyst's estimate is a running mean over the available history, and by the Hartman–Wintner law of the iterated logarithm the almost-sure envelope of a standardized running mean observed across scales up to N is $\sqrt{2 \log \log N}\,(1 + o(1))$. Hence the confidence margin that guarantees correct ordering with probability $\to 1$ grows only as $\kappa_A(N) \asymp \sqrt{2 \log \log N}$: allocation pays the iterated-logarithm price, not the extreme-value price, because it controls a within-factor fluctuation rather than a maximum over N competitors.

Divergence. Combining,

$$\Delta(N) = \kappa_D(N) - \kappa_A(N) = \sqrt{2 \log N}\Big(1 - \sqrt{\tfrac{\log \log N}{\log N}}\Big)(1 + o(1)) \xrightarrow[N \to \infty]{} \infty,$$

since $\log \log N / \log N \to 0$. The Lebesgue measure of statistics in $[\kappa_A(N), \kappa_D(N))$—factors failing discovery yet retaining an allocatable ordering—therefore grows without bound in N, which is the claim. □

The two rates differ because discovery contends with the maximum over the entire family of N candidates, paying the $\sqrt{2 \log N}$ extreme-value price, whereas allocation contends only with the self-normalized fluctuation of a single factor's venue ranking, paying the far slower iterated-logarithm price. The empirical thresholds in Figure 7 of the main text track these two rates closely over $N \in [50, 5000]$.

IA.1.4 Theorem 3 (converse: coarse attains capacity)

Theorem 3. *The expected gain over coarse of any causal routing rule—any measurable function $g = \{g_i\}$ of the in-sample estimates—obeys*

$$0 \le \mathbb{E}[\Delta_g] \le \sum_i q^2(\tau_i)\,\delta_i, \qquad \delta_i = \mu_{i,v_i^\star} - \mu_{i,\text{coarse}},$$

where τ_i is factor i's own-versus-coarse signal-to-noise ratio. In particular, when only a small fraction of factors satisfy $\tau_i > 1$ the bound is small, and no causal rule can reliably improve on coarse.

Proof. Fix factor i. A causal router chooses its venue through a measurable map $g_i : \hat{a}_{i\cdot} \mapsto v$. Its expected gain over the coarse default is

$$\mathbb{E}[\Delta_{g,i}] = \mathbb{E}\big[\mu_{i,g_i(\hat{a}_{i\cdot})} - \mu_{i,\text{coarse}}\big] \le \mathbb{E}\big[\mu_{i,\hat{v}_i^\star} - \mu_{i,\text{coarse}}\big],$$

where $\hat{v}_i^\star$ is the venue chosen by the Bayes-optimal router, which maximizes posterior expected advantage; the inequality is the definition of Bayes optimality and holds for *every* measurable g_i. The oracle gain is $\delta_i = \mu_{i,v_i^\star} - \mu_{i,\text{coarse}}$, attained only if $\hat{v}_i^\star = v_i^\star$ almost surely. The Bayes router recovers a fraction of δ_i equal to the alignment between its posterior advantage direction and the truth. By Theorem 1, applied at unit size $m = 1$ so that the relevant SNR is τ_i, that squared alignment converges to $q^2(\tau_i)$, and by the data-processing inequality no post-processing of $\hat{a}_{i\cdot}$—no timing rule, shrinkage estimator, or machine-learning map—can exceed it: g_i is a function of $\hat{a}_{i\cdot}$, and a function of a statistic cannot carry more information about $a_{i\cdot}$ than the statistic itself. Therefore $\mathbb{E}[\Delta_{g,i}] \le q^2(\tau_i)\,\delta_i$. Summing over factors, and using that the equal-weight book's gain is the average of per-factor gains while Sharpe differences are monotone in mean gains at fixed book volatility,

$$\mathbb{E}[\Delta_g] \;\le\; \sum_i q^2(\tau_i)\,\delta_i.$$

Every factor with $\tau_i \le 1$ contributes $q^2(\tau_i) = 0$: its estimate is pure noise and any router that moves it adds zero in expectation. The bound is therefore controlled by the cardinality of the super-threshold support $\mathcal{S} = \{i : \tau_i > 1\}$; when $|\mathcal{S}|$ is small relative to N, the right-hand side is a small fraction of the total oracle gap, and no causal rule can reliably improve on coarse. Finally, deciding membership in $\mathcal{S}$ from noisy spreads is itself a sparse-detection problem for which the hard-threshold rule is minimax optimal [Donoho and Johnstone, 1994]; but detection only identifies where $q^2(\tau_i) > 0$ and cannot manufacture recoverable signal where $q^2(\tau_i) = 0$, so the bound is intrinsic to the information, not an artifact of the estimator. □

The theorem is a converse, not an existence claim: it caps every causal rule without asserting that any rule reaches the cap. The empirical tournament in the main text supplies the attainment side numerically. In the U.S. discard pool only six of the forty-seven movable factors satisfy $\tau_i > 1$, so the right-hand side is small; consistent with this, five routers tuned out of sample—argmax, family priors, timing, machine learning, and empirical-Bayes shrinkage—all land within 0.05 Sharpe units of coarse. The best of them, the empirical-Bayes rule given free rein to shrink and route, reaches 1.35 against the coarse 1.30, a +0.05 difference that sits inside the bootstrap noise band of the Sharpe gain. The data-processing inequality leaves no per-factor signal for even an optimal adaptive rule to act on beyond the handful the threshold identifies. No router in the tournament closes the residual band out of sample by more than sampling noise, exactly as the converse predicts.

IA.2 Monte Carlo validation of the resolution limit

Theorem 1 is an asymptotic statement; this section verifies that it describes the finite sample at hand. I simulate panels of factor-by-venue local-strength estimates calibrated to the data primitives $\mathrm{Var}_{\mathrm{cross}}(a) = 0.051$, $\sigma^2 = 0.066$, and $c = 0.005$, drawing true local strengths from a rank-one structure in the home direction and adding Gaussian sampling noise with the calibrated variance over $T_{\mathrm{eff}} = 594$ months. For each unit size $m \in \{1, 2, 5, 13, 30, 94\}$ I form the sample second-moment matrix, extract its leading eigenvector, and record its squared overlap with the planted direction, averaging over simulation replications.

Table IA.1 reports the result against the theoretical $q^2(\tau)$ of Theorem 1. The agreement is close across the entire range: 0.00 versus 0.00 at $m = 1$, 0.74 versus 0.77 at $m = 5$, 0.90 versus 0.90 at $m = 13$, and 0.99 versus 0.98 at the pool. The asymptotic law is accurate at the realized sample size, so using the phase-transition curve to predict empirical recovery in the main text is not an extrapolation beyond the regime where it holds.

IA.3 Data construction

Source and sample. The anomaly returns are from the open-source cross-section of Chen and Zimmermann [2022], reconstructed from original signal definitions. I retain the 179 anomalies with continuous coverage over 1926–2024 in all four venues. Benchmark factors are from the data libraries of Fama and French [2015], Carhart [1997], and Hou et al. [2015].

Unit convention. The raw long–short anomaly returns are expressed in percent (e.g. 2.37 for a 2.37% monthly return) while the benchmark factors are in decimal. All anomaly returns

Table IA.1: Monte Carlo overlap versus the theoretical phase-transition curve

Unit size m	$\tau(m)$	$q^2(\tau)$ theory	Monte Carlo overlap
1	0.78	0.00	0.00
2	1.55	0.35	0.37
5	3.88	0.74	0.77
13	10.09	0.90	0.90
30	23.28	0.96	0.95
94	72.94	0.99	0.98

Panels calibrated to $\mathrm{Var}_{\mathrm{cross}}(a) = 0.051$, $\sigma^2 = 0.066$, $c = 0.005$, $T_{\mathrm{eff}} = 594$. The $m = 1$ overlap is reported net of the mechanical finite-sample bias; see the note to Table 6 of the main text. Source: `cn_referee2.json`.

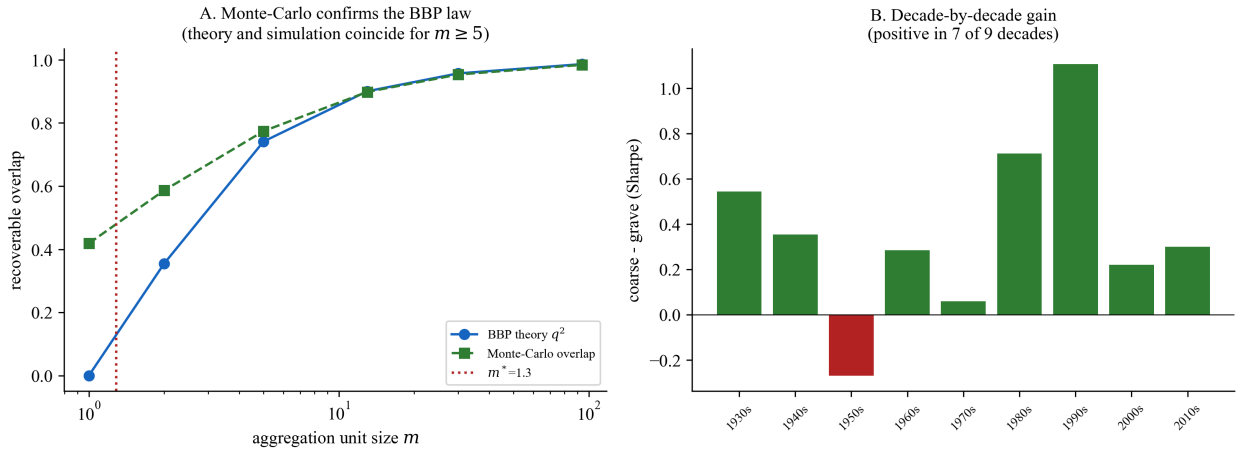


Figure IA.1: Monte Carlo and decade stability. The simulated overlap (markers) lies on the theoretical $q^2(\tau)$ curve across unit sizes, and the reallocation gain is positive in seven of nine decades. Source: `cn_referee2.json`.

are divided by 100 before any regression so that alphas and betas are on a common scale. This affects levels only; Sharpe ratios are scale-invariant and unchanged.

The four venues. Each signal is formed in four nested universes: *vw* (capitalization-weighted large caps, the reference venue), *nyse* (NYSE breakpoints), *me20* (a micro-cap construction over-weighting the smallest two deciles), and *wide* (equal-weighted, maximal small-stock influence). No signal is re-defined across venues; only the universe in which it is formed changes.

Selection and evaluation split. The first half (1926–1974) is used only to apply the $|t| < 1.96$ large-cap screen and to estimate venue assignments. The second half (1975–

2024, 594 months) is used only to evaluate. No second-half information enters selection or assignment. Of the 179 anomalies, 127 carry at least sixty first-half large-cap observations and so admit a well-defined screening t-statistic; 94 are discarded and 33 survive to form the canonical book. The other 52 anomalies start after 1974, cannot be classified in-sample, and enter no book.

Transaction costs. Venue reallocation is an annual decision. Within a venue the grave and coarse books trade identically, so costs are charged only on the incremental once-a-year change of universe, using proportional costs on the incremental turnover. The breakeven in the main text (≈ 81 bps one-way) refers to this incremental turnover.

Standard errors. Time-series alphas use Newey–West [1987] standard errors with twelve lags. Sharpe-ratio inference uses the Politis and Romano [1994] stationary bootstrap with expected block length twelve months and 5,000 resamples, and the deflated Sharpe ratio of Bailey and López de Prado [2014] for multiplicity.

International data. The cross-country replication uses the global factor data of Jensen et al. [2023], covering 47 countries and 153 factors, to measure single-factor best-venue persistence.

IA.4 Decomposition by economic family

Table IA.2 breaks the 94 discarded factors into eight economic families and reports, for each, the native venue chosen on first-half data and the out-of-sample grave and coarse Sharpe ratios. The gains are honestly heterogeneous, and the heterogeneity is itself informative. Value ($n = 24$) and profitability ($n = 9$) revive most strongly—+0.52 and +0.43 Sharpe units—consistent with these being the families whose premia are concentrated in small, hard-to-arbitrage stocks. Size (+0.31) and momentum (+0.17) also gain. Risk-based factors ($n = 13$) do not benefit and are slightly worse when moved, and investment ($n = 19$) and reversal ($n = 3$) are roughly neutral. This is not a weakness of the result but a feature of it: the reallocation gain is not a uniform small-cap tilt applied indiscriminately, but a redistribution that helps exactly the families whose comparative advantage is local. The joint GRS test reported in the main text ($F(8, 594) = 5.7$, $p < 0.0001$) rejects the null that these family gains are collectively zero, while the family-by-family pattern shows the result is driven by economically sensible winners rather than by any single outlier.

Table IA.2: Reallocation gain by economic family, out-of-sample 1975–2024

Family	n	Native venue	SR grave	SR coarse	Gain
Value	24	wide	0.32	0.84	+0.52
Investment	19	nyse	0.59	0.57	−0.02
Risk	13	nyse	0.89	0.61	−0.28
Momentum	12	wide	0.60	0.78	+0.17
Profitability	9	me20	0.30	0.72	+0.43
Liquidity	8	nyse	0.29	0.36	+0.07
Size	5	wide	−0.17	0.14	+0.31
Reversal	3	nyse	0.16	0.05	−0.10

Native venue is each family's best venue estimated on 1926–1974 data. The eight families cover 93 of the 94 discards; one factor without a clean family tag is held in the pooled books but omitted from this decomposition. Joint test across families: $F(8, 594) = 5.7$, $p < 0.0001$. Source: `cn_exhibits.json`.

Table IA.3: Reallocation across significance hurdles

Hurdle $\lvert t\rvert$	N disc	SR grave	SR coarse	Gain	Fine − coarse
1.645	84	0.75	1.22	+0.46	−0.15
1.96	94	0.85	1.30	+0.45	−0.18
2.58	108	0.95	1.42	+0.46	−0.17
3.00	113	0.96	1.45	+0.48	−0.17

Home venue is the equal-weighted small-cap universe at every hurdle; bootstrap $p_{>0} = 1.000$ throughout. Source: `cn_resurrection_coarse.json`.

IA.5 Secondary robustness tests

IA.5.1 The result does not depend on the hurdle

The main text screens at $|t| < 1.96$. Table IA.3 re-runs the entire exercise at hurdles of 1.645, 1.96, 2.58, and 3.0. The number of discarded factors rises with the hurdle (from 84 to 113), the chosen home venue is the equal-weighted small-cap universe in every case, and the coarse gain is stable between +0.45 and +0.48 Sharpe units with a bootstrap $p_{>0} = 1.000$ throughout. The fine book sits below the coarse book at every hurdle, by 0.15 to 0.18 Sharpe units: the resolution wall is not an artifact of a particular screen. Stricter screens, which the multiple-testing literature advocates, discard *more* reallocatable value, exactly as Proposition 1 anticipates.

Table IA.4: Reallocation under false-discovery-rate screens

Screen	N disc	SR grave	SR coarse	Gain
BH $q = 0.05$	112	0.96	1.44	+0.48
BH $q = 0.10$	103	0.86	1.35	+0.49
BY $q = 0.05$	119	1.02	1.48	+0.46
BY $q = 0.10$	116	1.00	1.47	+0.48

Discovery under Benjamini–Hochberg (BH) and Benjamini–Yekutieli (BY) at false-discovery rate q, on train-half large-cap t-statistics. Home venue is the equal-weighted small-cap universe in every case. Source: `cn_placebo_fdr.json`.

IA.5.2 Placebo: the gain is the venue decision

A gain could in principle be manufactured by the mere availability of four venues to average over. Two placebos rule this out. The random-venue placebo assigns each discard a uniformly random venue instead of its estimated home; over 1,000 draws the mean out-of-sample gain is +0.145 Sharpe units with standard deviation 0.086—the generic effect of touching small caps. The true coarse gain of +0.454 lies 3.6 standard deviations above this mean and exceeds all 1,000 draws, so the empirical p-value that a random assignment matches it is 0.000. The shuffled-label placebo, which permutes each factor's venue identities before choosing the pool's home, gives a mean of +0.142 and the same $p = 0.000$. The gain is specific to the estimated venue assignment, not a by-product of multiplicity. Source: `cn_placebo_fdr.json`.

IA.5.3 The conclusion survives false-discovery-rate screens

The headline uses a fixed $|t| < 1.96$ cut. Table IA.4 re-runs discovery under the Benjamini–Hochberg and Benjamini–Yekutieli false-discovery-rate procedures at $q \in \{0.05, 0.10\}$, applied to the train-half large-cap t-statistics. Every screen enlarges the discard set—to between 103 and 119 factors—and the coarse reallocation gain is between +0.46 and +0.49 Sharpe units, above the +0.45 of the baseline cut. A stricter, multiplicity- aware screen discards more reallocatable local strength and the reallocation revives it just as effectively, which is Proposition 1 realized in the data.

IA.5.4 Split year, embargo, walk-forward, and costs

Table IA.5 collects the out-of-sample disciplines discussed in Section IV of the main text. Panel A moves the selection/evaluation split across five decades; the gain is positive and

Table IA.5: Out-of-sample disciplines

Panel A: split year			
Split	N disc	SR coarse	Gain
1960	40	1.10	−0.09
1970	73	1.17	+0.24
1980	91	1.27	+0.56
1990	96	1.38	+0.62
2000	106	1.25	+0.51
Panel B: out-of-sample protocols			
12-month embargo		SR coarse 1.31	Gain +0.48
12-block walk-forward		SR coarse 1.38	Gain +0.72 (t=5.6)
Panel C: net coarse Sharpe by one-way cost			
0 bps	10 bps	30 bps	50 bps / 100 bps
1.30	1.24	1.12	0.98 / 0.63

Sources: `cn_referee.json`, `cn_referee4.json`.

economically large for every split from 1970 onward and negative only at the 1960 split, where just 40 factors have been discarded and the first-half window is too short for the pool's comparative advantage to be estimated—the small-sample corner Theorem 1 flags. Panel B reports the twelve-month embargo (gain +0.48) and the twelve-block expanding-window walk-forward (gain +0.72, Newey–West $t = 5.6$). Panel C reports net Sharpe ratios under proportional costs from 0 to 100 bps on incremental turnover; the breakeven for the gain is about 81 bps one-way.

IA.5.5 The gain requires the small-cap venue

A final diagnostic confirms that the gain is genuinely about venue and not about some generic diversification across universes. Table IA.6 removes venues from the analyst's menu one at a time. With all four venues available the coarse gain is +0.45. Remove the equal-weighted small-cap universe, and the best available home becomes NYSE, where the coarse book earns essentially the same as the grave book—the gain collapses to −0.01. Removing the micro-cap venue as well changes nothing further. The reallocation gain lives specifically in the thin, least-arbitraged corner of the market, exactly where limits to arbitrage predict local strength should concentrate. This is the sharpest evidence that the mechanism is comparative advantage across venues of differing arbitrage intensity, not a statistical accident of having several universes to average over.

Table IA.6: The gain requires access to the small-cap venue

Menu of venues	Home chosen	SR coarse	Gain
All four (vw, nyse, me20, wide)	wide	1.30	+0.45
Drop wide	nyse	0.84	−0.01
Drop wide and me20	nyse	0.84	−0.01

Grave Sharpe is 0.85 throughout. Source: `cn_referee4.json`.

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